

let $y = uv$ $x=1$ $f(x) = -x$
 $f(1) = -1$

B.Sc (HONS.) IN CSE, FIRST YEAR, FIRST SEMESTER EXAMINATION, 2020

CALCULUS

[According to the New Syllabus]

Subject Code : CSE-510205

Examination Code : 5611

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks of the questions. Answer any four questions.]

Group A—Differential Calculus

- | | Marks |
|---|-------|
| 1. (a) Define function, even function and odd function. | 4 |
| (b) Draw a graph of the following function : | 8 |

$$f(x) = \begin{cases} x^2 + 1 & \text{when } x < 0 \\ x & \text{when } 0 \leq x \leq 1 \\ \frac{1}{x} & \text{when } 1 < x \end{cases}$$

Find the domain and range of the above function.

- | | |
|---|---|
| (c) Define continuity. Discuss the continuity and differentiability of the following function at $x = 1$ when | 8 |
|---|---|

$$f(x) = \begin{cases} -x & \text{when } -2 \leq x \leq 0 \\ x & \text{when } 0 < x < 1 \\ 3-x & \text{when } 1 \leq x \leq 2 \end{cases}$$

2. (a) Find $\frac{dy}{dx}$ (any three) :

- $x^y + y^x = a^b$
- $y = (\sin x)^{\ln x} + \sin^2(\cos^{-1} x)$
- $y = \sin^{-1} \left(\frac{a + b \cos x}{b + a \cos x} \right)$
- $y = \tan^{-1} \left(\frac{1 + \tan x}{1 - \tan x} \right)$

- (b) State Leibnitz theorem. If $y = \sin(m \sin^{-1} x)$ then prove that, $(1 - x^2)y_{n+2} - (2x + 1)xy_{n+1} + (m^2 - n^2)y_n = 0$.

[Please turn over]

3. (a) State and prove Cauchy's mean value theorem. Marks: 8
 (b) Verify mean value theorem for the function $f(x) = 3 + 2x - x^2$ in the interval $(0, 1)$. 6
 (c) Prove that x^x has a minimum value at $x = \frac{1}{e}$. 6

Group B—Integral Calculus

4. (a) Evaluate (any five):

4×5=20

- (i) $\int \frac{e^x(1+x)dx}{\cos^2(xe^x)}$
 (ii) $\int \sqrt{1 + \sec x} dx$
 (iii) $\int \frac{dx}{(x^2+1)\sqrt{x^2+4}}$
 (iv) $\int \tan^5 x dx$
 (v) $\int \cos 2 \cot^{-1} \sqrt{\frac{1-x}{1+x}} dx$
 (vi) $\int \frac{dx}{\sqrt{(x-\alpha)(\beta-x)}}$

5. (a) State and prove Fundamental theorem of integral calculus. 8
 (b) Evaluate: $\lim_{n \rightarrow \infty} \frac{1}{n} [(n+1)(n+2)(n+3) \dots (n+n)]^{1/n}$. 6
 (c) If $I_n = \int_0^{\pi/4} \tan^n \theta d\theta$ then show that $n(I_{n+1} + I_{n-1}) = 1$. 6
6. (a) Find the length of the perimeter of the Cardoid $r = 8(1 + \cos \theta)$. 6
 (b) Find the whole area of the hypocycloid $x^{2/3} + y^{2/3} = a^{2/3}$. 7
 (c) Find the area of a loop of the curve $a^2 y^2 = x^2 (a^2 - x^2)$. 7